

EXAM IN MULTIVARIATE METHODS ST3101

1 November 2024

Time: 5 hours

Allowed aids: Pocket calculator, language dictionary

The exam consists of five questions. To score maximum points on a question solutions need to be clear, detailed and well motivated.

GOOD LUCK!

Formula sheet on pages 12 and forward

Question 1. (16 Points)

- (a) The Common Brushtail Possum (*Trichosurus vulpecula*¹), is a nocturnal marsupial native to Australia. Figure 1 is a digital image of a Common Brushtail Possum. Panels (b), (c), and (d) of Figure 1, are the 700×700 representations of the red, green, and blue colors, or *channels*. Each pixel takes a continuous value between 0 and 1 on each of $x_{ij}^{(red)}$, $x_{ij}^{(green)}$, and $x_{ij}^{(blue)}$. The black and white image, panel (a), has been constructed by letting each black and white pixel be $x_{ij}^{(bw)} = 0.299x_{ij}^{(red)} + 0.587x_{ij}^{(green)} + 0.114x_{ij}^{(blue)}$. Explain briefly how you would represent data and proceed if you wanted to apply one of the methods of ST3101. Consider either *data reduction* or *classification* (the latter assuming, for example, a sample of possum pictures). Be creative but brief.
- (b) Let $\mathbf{x}$ be a 3×1 vector of random variables, and

$$\mathbf{z} = \mathbf{A}\mathbf{x}$$

for

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{4} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

Assuming that $\mathbf{z} \sim N_3(\mathbf{0}, \mathbf{I})$, what distribution does $\mathbf{x}$ follow? Provide as detailed a characterisation as you can muster.

- (c) What is the determinant of

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

¹From the Greek for ‘furry tailed’ and the Latin for ‘little fox’.

and is $\mathbf{A}$ invertible? Qualified and well motivated guesses may warrant points. It might be helpful to recall how the determinant is related to Gauss-Jordan elimination.

Question 2. (21 Points)

Table 1 summarises data for $n = 104$ brushtail possums, trapped at seven Australian sites from Southern Victoria to central Queensland.

	Mean	Sd	Questions
site			a coded categorical variable stating where the possum was caught (1-7)
Pop	0.44		a dichotomous variable indicating if the possum is from Victoria (1) or elsewhere (0)
sex	0.41		a dichotomous variable stating the sex of the possum (1 for female)
age	3.83	1.89	a numerical discrete variable stating the age of the possum (1-9)
hdlngth	92.60	3.57	a numerical continuous variable stating the head length of the possum in mm
skullw	56.88	3.11	a numerical continuous variable stating the skull width of the possum in mm
totlngth	87.09	4.31	a numerical continuous variable stating the total length of the possum in cm
tail	37.01	1.96	a numerical continuous variable stating the tail length in cm
footlght	68.46	4.37	a numerical continuous variable stating the foot length in mm
earconch	48.13	4.11	a numerical continuous variable stating the ear conch length in mm
eye	15.05	1.05	a numerical continuous variable stating the distance from medial canthus to lateral canthus of right eye in mm (it stands for the eye size)
chest	27.00	2.05	a numerical continuous variable stating the chest girth in cm
belly	32.59	2.76	a numerical continuous variable stating the belly girth in cm

Table 1: Data for $n = 104$ Brushtail Possums caught in Australia

The correlation matrix for the 10 numerical variables, in order **age**, **hdlngth**, ..., **belly** (it might be helpful to number the variables in the margin of Table 1 and/or note down the



(a)



(b)



(c)



(d)

Figure 1: Common Brushtail Possum (*Trichosurus vulpecula*). Image consists of 700×700 pixels, in black and white (a), red component (or channel) (b), green component (c), and blue component (d)

variable names next to their rows in $\mathbf{R}$), is given by

$$\mathbf{R} = \begin{bmatrix} 1.000 & 0.312 & 0.280 & 0.251 & 0.117 & 0.125 & 0.053 & 0.235 & 0.325 & 0.346 \\ 0.312 & 1.000 & 0.711 & 0.691 & 0.287 & 0.389 & 0.121 & 0.347 & 0.631 & 0.563 \\ 0.280 & 0.711 & 1.000 & 0.526 & 0.256 & 0.275 & -0.001 & 0.322 & 0.630 & 0.452 \\ 0.251 & 0.691 & 0.526 & 1.000 & 0.566 & 0.443 & 0.154 & 0.248 & 0.578 & 0.519 \\ 0.117 & 0.287 & 0.256 & 0.566 & 1.000 & -0.126 & -0.385 & 0.198 & 0.175 & 0.294 \\ 0.125 & 0.389 & 0.275 & 0.443 & -0.126 & 1.000 & 0.769 & 0.005 & 0.451 & 0.302 \\ 0.053 & 0.121 & -0.001 & 0.154 & -0.385 & 0.769 & 1.000 & -0.165 & 0.199 & 0.067 \\ 0.235 & 0.347 & 0.322 & 0.248 & 0.198 & 0.005 & -0.165 & 1.000 & 0.149 & 0.236 \\ 0.325 & 0.631 & 0.630 & 0.578 & 0.175 & 0.451 & 0.199 & 0.149 & 1.000 & 0.606 \\ 0.346 & 0.563 & 0.452 & 0.519 & 0.294 & 0.302 & 0.067 & 0.236 & 0.606 & 1.000 \end{bmatrix}$$

The eigenvalues λ of $\mathbf{R}$, and the matrix $\mathbf{W}_{1:6}$ with the first 6 eigenvectors for $\mathbf{R}$ as columns, are

$$\lambda = \begin{bmatrix} 4.107 \\ 1.966 \\ 0.978 \\ 0.792 \\ 0.721 \\ 0.530 \\ 0.319 \\ 0.267 \\ 0.163 \\ 0.157 \end{bmatrix}, \quad \mathbf{W}_{1:6} = \begin{bmatrix} -0.228 & 0.056 & 0.605 & 0.610 & 0.324 & 0.312 \\ -0.426 & 0.034 & 0.004 & -0.175 & -0.164 & 0.167 \\ -0.382 & 0.094 & 0.079 & -0.194 & -0.463 & 0.403 \\ -0.409 & 0.052 & -0.358 & -0.036 & 0.302 & 0.132 \\ -0.199 & 0.453 & -0.483 & 0.123 & 0.412 & 0.085 \\ -0.267 & -0.527 & -0.103 & -0.133 & 0.233 & -0.004 \\ -0.105 & -0.646 & -0.006 & -0.063 & 0.263 & 0.024 \\ -0.192 & 0.269 & 0.504 & -0.647 & 0.341 & -0.228 \\ -0.401 & -0.090 & -0.011 & 0.159 & -0.379 & -0.085 \\ -0.366 & 0.050 & 0.025 & 0.280 & -0.098 & -0.793 \end{bmatrix}$$

Assume the model $\mathbf{y} = \mathbf{W}^\top \mathbf{x}$, where $\mathbf{x}$ is the 10×1 vector of measurements for a random possum. Use reported quantities as estimates of the true model parameters (keep in mind the assumptions made about $\mathbf{x}$ implied by the eigen-decomposition of the correlation matrix $\mathbf{R}$; take into account any centering or scaling that has been done).

- What is the variance-covariance matrix $V(\mathbf{y})$?
- What is $\text{corr}(Y_1, Y_{10})$?
- What proportion of variance $V(Y_2)$ out of $\text{tr}(V(\mathbf{y}))$?
- What variable has the strongest correlation with Y_2 ?
- Which of $Y_1, \dots, Y_6$ has the strongest correlation with the length of a possums tail?
- What is the predicted y_1 for a possum that has an unstandardised chest girth of 28cm and a belly girth of 34cm, and all other measurements equal to their mean?

Question 3. (20 Points)

Factor analysis is performed on $\mathbf{R}$ (as provided in Question 2). An unrotated Principal axes solution yields the loadings matrix

$$\tilde{\mathbf{\Lambda}} = \begin{bmatrix} 0.38 & -0.06 & -0.16 \\ 0.84 & -0.07 & -0.15 \\ 0.73 & -0.14 & -0.26 \\ 0.84 & -0.11 & 0.39 \\ 0.39 & -0.60 & 0.45 \\ 0.55 & 0.69 & 0.15 \\ 0.23 & 0.89 & 0.13 \\ 0.32 & -0.25 & -0.15 \\ 0.77 & 0.07 & -0.17 \\ 0.67 & -0.08 & -0.09 \end{bmatrix}.$$

With varimax rotation and rotation matrix

$$\mathbf{Q} = \begin{bmatrix} 0.839 & -0.166 & -0.518 \\ 0.347 & 0.897 & 0.275 \\ 0.419 & -0.411 & 0.810 \end{bmatrix}$$

the following three-factor loadings matrix is obtained (with λ_{11} missing)

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_{11} & 0.04 & 0.05 \\ 0.80 & 0.19 & 0.27 \\ 0.77 & 0.05 & 0.15 \\ 0.53 & 0.30 & 0.71 \\ 0.19 & -0.28 & 0.77 \\ 0.27 & 0.85 & 0.07 \\ -0.02 & 0.92 & -0.16 \\ 0.38 & -0.15 & 0.12 \\ 0.73 & 0.29 & 0.15 \\ 0.62 & 0.14 & 0.24 \end{bmatrix}$$

In the following, consider the rotated solution, $\mathbf{\Lambda}$.

- (a) Find λ_{11} .
- (b) Draw the model diagram, omitting loadings that are smaller than $\frac{1}{2}$ in absolute value (indicate sign but you do not need to write down the actual numbers).
- (c) Briefly, try to interpret the factors based upon your diagram and with reference to the variable definitions in Table 1.
- (d) What is the estimated correlation between chest girth and a belly girth?
- (e) Provide an estimate of the variance of ϵ_1 .

Question 4. (16 Points)

In the sample of possums in Table 1, $n_1 = 46$ animals have been captured in the state of Victoria, and $n_0 = 58$ have been captured in another Australian state or territory.² Figure 2 plots the first two principal components of the (scaled) numerical data in Table 1, with observations marked by site.

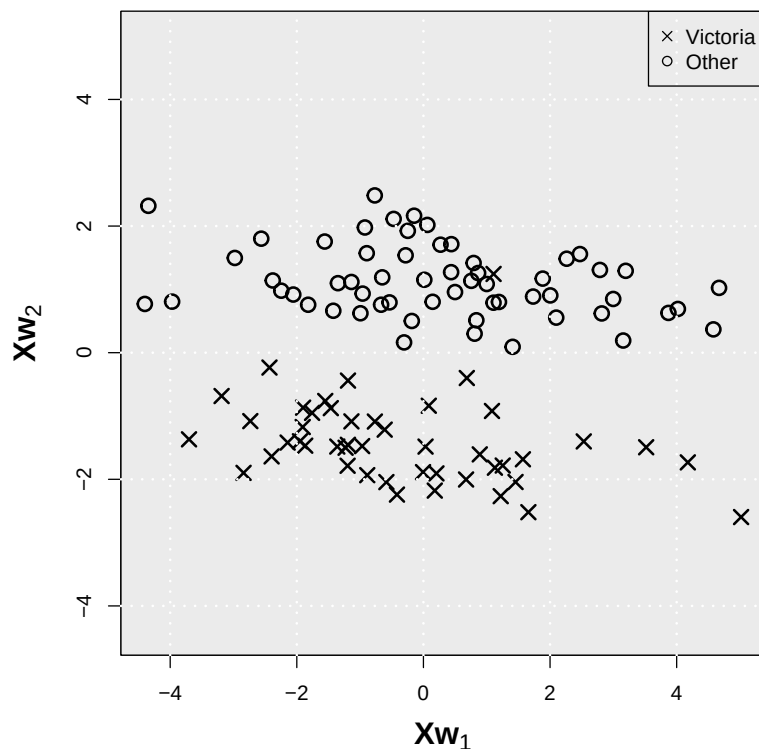


Figure 2: The first two principal components for the sample of $n = 104$ Common Brushtail Possums in Table 1 with plot symbol indicating whether Possum was captured in Victoria or in another Australian State or Territory. Note the data matrices, $\mathbf{X}$, here have been standardised

To try to discriminate and classify possums, an expert on marsupials has made a number of calculations based on the sample of Table 1. Some of these calculations are reported below.

Some differences between Victorian and other possums in the sample are in tail and foot length. The average tail length in Victoria is 35.93, and in the other locations it is 37.86. The average foot length in Victoria is 72.31, and in the other locations it is 65.40.

²The states of Australia are Victoria, New South Wales, South Australia, Western Australia, Queensland, and Tasmania. The two territories are the Northern Territory and the Australian Capital Territory.

Doing a logistic regression of site (Victoria: 1, and Other: 0), on a selection of unscaled and uncentered variables yields the results in Table 2.

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	42.24	12.28	3.44	0.00
sexTRUE	0.92	0.73	1.26	0.21
age	0.12	0.18	0.65	0.51
hdlngth	-0.25	0.17	-1.53	0.13
skullw	-0.31	0.15	-2.02	0.04
totlngth	0.70	0.18	3.94	0.00
taill	-1.95	0.40	-4.82	0.00
eye	-0.05	0.33	-0.16	0.87
chest	0.55	0.28	2.01	0.04
belly	-0.15	0.14	-1.09	0.27

Table 2: Logistic regression of Victorian or other, on a subset of the for $n = 104$ Brushtail Possums in Table 1

Furthermore, for the subset of (unscaled and uncentered) variables **hdlngth**, **skullw**, **footlngth**, **earconch**, and **eye** (in this order), the following sample variance covariance matrices are provided

$$\mathbf{S}_0 = \begin{bmatrix} 17.17 & 10.91 & 7.51 & 1.74 & 1.48 \\ 10.91 & 12.26 & 6.13 & 0.98 & 1.28 \\ 7.51 & 6.13 & 7.67 & 0.84 & 0.74 \\ 1.74 & 0.98 & 0.84 & 2.49 & -0.18 \\ 1.48 & 1.28 & 0.74 & -0.18 & 1.47 \end{bmatrix}$$

and

$$\mathbf{S}_1 = \begin{bmatrix} 7.48 & 4.28 & 4.44 & 1.92 & 1.11 \\ 4.28 & 6.56 & 2.43 & 0.45 & 0.71 \\ 4.44 & 2.43 & 6.86 & 1.92 & 0.35 \\ 1.92 & 0.45 & 1.92 & 5.34 & -0.10 \\ 1.11 & 0.71 & 0.35 & -0.10 & 0.61 \end{bmatrix}.$$

The inverse of the pooled sample covariance matrices is

$$\mathbf{S}_{\text{pool}}^{-1} = \begin{bmatrix} 0.18 & -0.10 & -0.08 & -0.05 & -0.09 \\ -0.10 & 0.19 & -0.03 & 0.02 & -0.04 \\ -0.08 & -0.03 & 0.21 & -0.03 & 0.01 \\ -0.05 & 0.02 & -0.03 & 0.26 & 0.09 \\ -0.09 & -0.04 & 0.01 & 0.09 & 0.95 \end{bmatrix}$$

Finally, the mean vectors and the (partial) linear discriminant are, respectively,

$$\bar{\mathbf{x}}_0 = \begin{bmatrix} 92.61 \\ 57.07 \\ 65.40 \\ 44.91 \\ 15.18 \end{bmatrix}, \quad \bar{\mathbf{x}}_1 = \begin{bmatrix} 92.60 \\ 56.65 \\ 72.31 \\ 52.19 \\ 14.87 \end{bmatrix}, \quad \text{and} \quad \boldsymbol{\gamma} = \begin{bmatrix} 0.78 \\ 0.13 \\ \gamma_3 \\ -1.68 \\ -0.41 \end{bmatrix}.$$

The possum in Figure 1, has been measured and found to have measurements as in Table 3.

sex	0
age	8
hdlngth	94
skullw	60
totlngth	89
taill	36
footlght	74
earconch	54
eye	15
chest	28
belly	36

Table 3: Measures for the possum in Figure 1

- (a) Given the tail and foot lengths, how would you explain the pattern in Figure 2?
- (b) How would you classify the possum in Figure 1 using linear discrimination analysis?
- (c) What assumption (-s) would you require for the linear discriminant to have the desired classification probabilities? Provide a sketch of the relevant test for the variance of the variables, indicating what above quantities you would require.
- (d) How would you classify the possum in Figure 1 using logistic regression?

Question 5. (7 Points)

Consider the images in Figures 3, and 4.

- (a) What value k has been used in Figure 4 and Figure 3?
- (b) If each unique colour requires 64 bits, how many bits does the original RBG image in Figure 1, panels (b), (c), and (d) require, compared to the (colour version of the) k -means image in Figure 4?
- (c) How many how many bits does the original black and white image in Figure 1(a) require compared to the k -means image in Figure 3?
- (d) For Figure 3(b), what is greater,

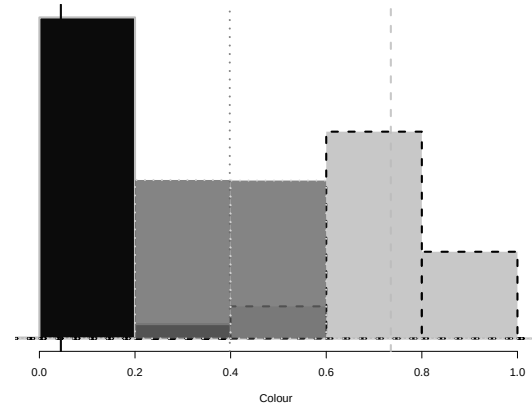
$$SS_w = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_i^{(j)} - \bar{x}^{(j)})^2,$$

or

$$SS_T = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_i^{(j)} - \bar{x})^2,$$



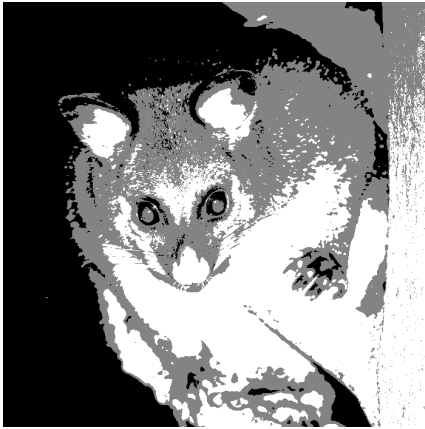
(a)



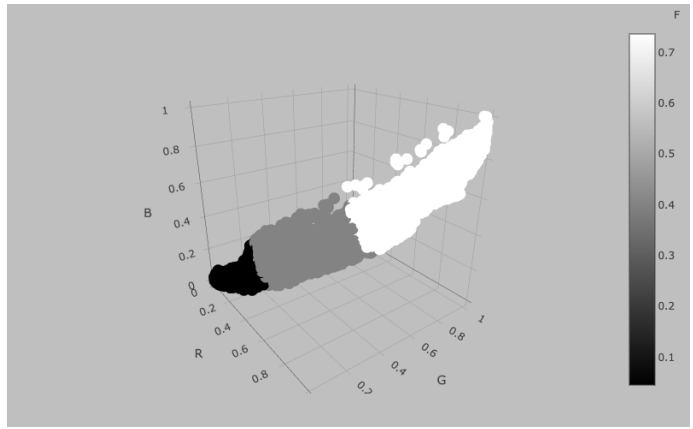
(b)

Figure 3: K -means clustering of the Figure 1(a). Panel (b) gives the original data with cluster means and panel (a), the cluster means arranged as and 700×700 pixel image

where $x_i^{(j)}$ is the i th observation in cluster j , $\bar{x}^{(j)}$ is the cluster mean of cluster $j = 1, \dots, k$, and $\bar{x}$ is the grand mean.



(a)



(b)

Figure 4: K -means clustering of the RGB matrices in Figure 1(b),(c), and (d). Panel (b) gives the original data with cluster means and panel (a), the cluster means arranged as and 700×700 pixel image, where each pixel has been transformed to grey scale according to the formula $0.299Red + 0.587Green + 0.114Blue$

Formula Sheet for the Exam in Multivariate Methods

Vectors and matrices

- Length of a vector $\mathbf{a} = (a_1, a_2, \dots, a_p)$

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_p^2}$$

- The trace of a square $p \times p$ matrix $\mathbf{A}$ is given by

$$\text{tr}(\mathbf{A}) = \sum_{i=1}^p a_{ii}$$

- Determinant of a 2×2 matrix $\mathbf{A}$

$$\det(\mathbf{A}) = |\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}$$

- Inverse of a 2×2 matrix $\mathbf{A}$

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{pmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{pmatrix}$$

- Eigenvalues are the roots of the characteristic equation

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

For each eigenvalue the solution to

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{x} = \mathbf{0}$$

gives the associated eigenvector $\mathbf{x}$

Distances

- Euclidean

$$D_{ik} = \sqrt{\sum_{j=1}^p (x_{ij} - x_{kj})^2}$$

- Statistical

$$SD_{ik} = \sqrt{\sum_{j=1}^p \left(\frac{x_{ij} - x_{kj}}{s_j} \right)^2}$$

- Mahalanobis

$$MD_{ik} = \sqrt{(\mathbf{x}_i - \mathbf{x}_k)^T \mathbf{S}^{-1} (\mathbf{x}_i - \mathbf{x}_k)}$$

For $p = 2$

$$MD_{ik} = \sqrt{\frac{1}{1 - r^2} \left[\frac{(x_{i1} - x_{k1})^2}{s_1^2} + \frac{(x_{i2} - x_{k2})^2}{s_2^2} - \frac{2r (x_{i1} - x_{k1})(x_{i2} - x_{k2})}{s_1 s_2} \right]}$$

Mean-correction and covariance

- Mean-corrected data

$$\mathbf{X}_m = \{x_{ij}\}_{(n \times p)} = \{X_{ij} - \bar{X}_j\}$$

- Covariance

$$\mathbf{S}_{(p \times p)} = \{s_{ij}\} = \left\{ \frac{\sum_{i=1}^n x_{ij} x_{ik}}{n - 1} \right\} = \frac{\mathbf{SSCP}}{df} = \frac{1}{n - 1} \mathbf{X}_m^T \mathbf{X}_m$$

Group Analysis

- Total sum of squares and cross products

$$\mathbf{SSCP}_{\text{total}} = \mathbf{SSCP}_{\text{within}} + \mathbf{SSCP}_{\text{between}}$$

- Pooled within-group sum of squares and cross products

$$\mathbf{SSCP}_{\text{within}} = \sum_{\ell=1}^g \mathbf{SSCP}_{\ell}$$

- Pooled covariance matrix

$$\mathbf{S}_{\text{pooled}} = \frac{\mathbf{SSCP}_{\text{within}}}{n - g}$$

- Between-group sum of squares and cross products

$$\mathbf{SSCP}_{\text{between}} = \mathbf{SSCP}_{\text{total}} - \mathbf{SSCP}_{\text{within}}$$

For $g = 2$ groups

$$\mathbf{SSCP}_{\text{between}} = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^T$$

Factor Analysis

- For the two-factor model

$$\text{Var}(x) = \lambda_1^2 + \lambda_2^2 + \text{Var}(\epsilon) + 2\lambda_1\lambda_2\phi$$

$$\text{Cor}(x, \xi_1) = \lambda_1 + \lambda_2\phi$$

$$\text{Cor}(x_j, x_k) = \lambda_{j1}\lambda_{k1} + \lambda_{j2}\lambda_{k2} + (\lambda_{j1}\lambda_{k2} + \lambda_{j2}\lambda_{k1})\phi$$

- RMSR for EFA

$$RMSR = \sqrt{\frac{\sum_{i=1}^p \sum_{j=i+1}^p res_{ij}^2}{p(p-1)/2}}$$

- RMSR for CFA

$$RMSR = \sqrt{\frac{\sum_{i=1}^p \sum_{j=i}^p (s_{ij} - \hat{\sigma}_{ij})^2}{p(p+1)/2}}$$

Two-Group Discriminant Analysis

- Maximize

$$\lambda = \frac{\gamma^T \mathbf{B} \gamma}{\gamma^T \mathbf{W} \gamma}$$

- Fisher's linear discriminant function

$$\gamma^T = (\mu_1 - \mu_2)^T \Sigma^{-1}$$

- Wilks' Λ

$$\Lambda = \frac{|\mathbf{SSCP}_w|}{|\mathbf{SSCP}_t|}$$

$$F = \left(\frac{1 - \Lambda}{\Lambda} \right) \left(\frac{n_1 + n_2 - p - 1}{p} \right) \sim F(p, n_1 + n_2 - p - 1)$$

- Classification based on decision theory: assign the observation to group 1 if

$$Z \geq \frac{\bar{Z}_1 + \bar{Z}_2}{2} + \ln \left[\frac{p_2 C(1|2)}{p_1 C(2|1)} \right]$$

Logistic regression

- Odds of the event $Y = 1$

$$odds = \frac{p}{1-p}$$

where

$$p = P(Y = 1)$$

- Probability of the event $Y = 1$ as a function of the explanatory variables

$$P(Y = 1|X_1, X_2, \dots, X_k) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k)}}$$

Quadratic equation

- The roots of the quadratic equation $ax^2 + bx + c$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$